JIBRAN GUL KAYANI

Hostel City, Islamabad | 0309-9991997 | jibrankayani776@gmail.com
linkedin.com/in/jibran-gul-kayani | portfolio-8r62.vercel.app

Motivated Hardware and PCB Design Engineer with hands-on experience in schematic design, PCB layout, and hardware validation. Skilled in Altium Designer, KiCad, Proteus, and LTspice, with practical exposure to sensor interfacing and power electronics. Strong collaborator with experience supporting real-world product development and testing.

WORK EXPERIENCE

Integrated Technologies – Hardware Design Engineer Aug 2025 – Present

- Assist in hardware design and testing for environmental monitoring systems.
- Perform real-world testing to validate sensor readings and overall device performance.
- Identify, debug, and resolve hardware issues during development stages.
- Support validation procedures and contribute to efficient testing workflows.

Electronic Interconnect Engineering – Intern Mar 2025 – Aug 2025

- Assisted in firmware testing and sensor interfacing using Arduino and ESP32.
- Learned PCB schematic design, layout routing, and DRC in Altium Designer.
- Performed SMD and through-hole soldering and assisted in board assembly.
- Supported debugging and basic hardware validation activities.

SKILLS

Design Tools: Altium Designer, EasyEDA, KiCad
Simulation Tools: LTspice, Proteus
Technical Skills: Schematic Capture, Circuit Design, Multilayer PCB Design, Power Electronics PCB, PCB Stack-up Design, Fabrication Files, PCB Debugging & Troubleshooting, Hardware Prototyping & Testing, SMD / Through-hole Soldering,

PROJECTS

Mixed-Signal Central Controller Board

- Designed schematics and PCB layout for a mixed-signal controller board.
- Integrated o STM32F407, communication interfaces(Ethernet, USB 2.0, CAN, UART), analog sensing, motor control, and audio circuitry.
- Focused on power distribution, signal integrity, and board-level validation.

Medical Monitoring System

- Developed a multi-sensor medical monitoring system using STM32L496ZET6.
- Integrated modules for electrocardiogram, heart rate, blood oxygen saturation (SpO2) and ambient temperature data and implemented real-time data acquisition.
- Prepared schematics and manufacturing files for PCB fabrication.

Sensor Board

- Designed a dedicated sensor board consolidating multiple environmental sensors.
- Integrated CO_2, CO, LPG gas sensors, and ESP-CAM module.
- Performed hardware testing and validation of the PCB board.

Smart Home Monitoring System

- Designed and developed an ESP32-based smart home monitoring system.
- Integrated air quality, temperature, and humidity sensors.
- Implemented reliable sensor interfacing and real-time monitoring.

Multi-Peripheral Control Board

- Designed a power and control PCB supporting multiple peripherals.
- Implemented voltage regulation, protection, and power distribution circuits.
- Optimized PCB layout for stable operation and testing.

Final Year Project (FYP) – Soil Moisture Monitoring System

- Designed a solar-powered soil moisture monitoring system.
- Implemented wireless communication and cloud-based monitoring.
- Developed a mobile application for remote data visualization.

EDUCATION

BS Electrical Engineering Sep 2019 – Jan 2025
COMSATS University Islamabad

F.Sc Pre-Engineering Sep 2017 – Aug 2019
Govt. Postgraduate College Jhelum

Matriculation Jun 2017
F.G Boys Public School Jhelum Cantt

LANGUAGES

Urdu: Native
English: Professional Working Proficiency

CERTIFICATES

Registered Engineer — Pakistan Engineering Council (PEC)